

Title

Automated Student Attendance System Using Real-Time Face Recognition with Entry-Exit Validation

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Abstract

Conventional attendance practices, such as calling out names or circulating a sheet, consume several minutes of class time and are easily abused through proxy marking. This work proposes a fully automated attendance system that leverages real-time face recognition and a dual-camera setup at the classroom entrance and exit. The system acquires facial images at both doors, generates 128-dimensional embeddings using a pre-trained convolutional neural network from the dlib library, and compares them with stored encodings via Euclidean distance. A distance threshold of 0.5 is applied so that only highly reliable matches are accepted, thereby reducing false positives. A student is counted as present only when their face is confidently identified at both entry and exit, and the time gap between these events is at least 40 minutes. Attendance logs are automatically exported to a CSV file containing the student's name, date, entry time, exit time, duration of stay, and final status. Experiments show that the distance threshold lowers incorrect matches, and the two-camera design effectively blocks proxy attendance and short-duration visits from being recorded as full presence.

1. Introduction

Accurate attendance tracking is an essential administrative task in schools and universities. In typical classrooms, instructors either call the roll or pass around a sign-in sheet at the beginning of the session. This procedure can consume 5–10 minutes of each lecture and is vulnerable to misuse when one student marks another as present despite their absence. As the number of students in a class grows, these manual methods become increasingly inefficient and unreliable.

Biometric approaches have been explored to automate attendance, with facial recognition standing out as a convenient option because it works with standard cameras and does not require physical contact or tokens. However, common single-camera facial attendance systems primarily mitigate the proxy problem and still have serious limitations. They usually cannot confirm how long a student actually stays in the classroom; a student might walk in, get recognized, and leave after a couple of minutes while still being marked present for the entire session. Moreover, systems that rely solely on Boolean outputs from face recognition libraries, without inspecting the underlying distance values, may accept borderline matches and thus introduce false positives into the record.

The system proposed in this paper is designed to tackle these three issues simultaneously. It removes manual intervention by performing real-time automated recognition, blocks proxy

attendance through biometric verification at entry, measures the true duration of presence using a dual entry and exit camera design, and filters out uncertain identifications using a Euclidean distance threshold of 0.5. The paper is organized as follows: Section 2 surveys related literature, Section 3 states the research gap, Section 4 formulates the problem, Section 5 explains the methodology, Section 6 outlines the system workflow, Section 7 presents and analyzes results, Section 8 concludes the work, and Section 9 describes possible future extensions.

2. Literature Review

Viola and Jones (2001) introduced a fast face detection method based on Haar cascade classifiers, which remains popular for real-time applications due to its computational efficiency. Their sliding-window framework with boosted Haar-like features underpins the detection stage in this project through OpenCV's `haarcascade_frontalface_default.xml` model.

King (2009) presented the dlib library, a C++ machine learning toolkit that includes a highly accurate face recognition model trained using deep metric learning. In dlib, each detected face is represented as a 128-dimensional feature vector, and the Euclidean distance between two vectors serves as a similarity measure, which is the core matching mechanism adopted in this system.

Schroff et al. (2015) proposed FaceNet, which maps facial images into a compact Euclidean space where distances correspond directly to identity similarity. Using triplet loss, they demonstrated that 128-dimensional embeddings are sufficient for reliable face verification, providing support for the embedding size chosen in this work.

Several prior implementations have used OpenCV-based face recognition for attendance automation. A recurring limitation in these designs is the use of only one camera at the entrance, which can confirm that a student passed through the door but cannot guarantee that they remained for the full duration of the class. In addition, many systems rely on the default Boolean threshold in `compare_faces()` instead of examining the raw Euclidean distances, which makes them prone to accepting near matches as correct identifications.

RFID-based attendance methods also exist and deliver reliable identification by scanning unique tags assigned to each student. However, they require extra hardware tokens, which add cost and can be forgotten, lost, or handed to someone else, thereby reintroducing the proxy attendance problem.

3. Research Gap

Current facial recognition attendance systems largely succeed in removing manual roll calls but still leave two important issues unresolved. First, typical implementations do not validate how long the recognized student actually stays in the room; detection at entry alone is usually enough to be marked present. Second, systems that depend on thresholded Boolean outputs from recognition libraries without directly checking distance scores cannot reliably distinguish strong matches from borderline ones, increasing the risk of false positives.

The system presented here addresses both shortcomings. It introduces a Euclidean distance-based confidence cutoff of 0.5 and employs a dual-camera architecture to measure the time spent in the classroom before finalizing attendance. Only when a student is confidently identified at both cameras and meets the minimum duration criterion are they counted as present.

4. Problem Formulation

Consider a set of N registered students, each represented by multiple pre-computed 128-dimensional facial embeddings stored in a serialized file. The objective of the system is to perform the following operations.

- Detect and recognize every student's face at the entry camera, accepting a match only when the Euclidean distance to a stored embedding is below 0.5.
- Log the entry timestamp for each successfully recognized student.
- Detect and recognize the same student at the exit camera using the same distance threshold.
- Record the exit timestamp for each confirmed match at the exit camera.
- Compute the time spent in the classroom as the difference between exit and entry times in minutes.
- Assign the label "Present" if and only if both entry and exit events are confirmed and the duration is at least 40 minutes; otherwise, mark the student as "Absent."

Students who are not recognized by either camera or recognized only once are also classified as absent.

5. Methodology

5.1 Data Collection

For each enrolled student, video clips are recorded under typical classroom lighting conditions. The script `v2p_10.py` opens each video using `cv2.VideoCapture()`, reads the frame rate via `cap.get(cv2.CAP_PROP_FPS)`, and selects a sampling interval as `max(int(fps/10), 1)` so that roughly ten frames per second are extracted regardless of the original frame rate. The sampled frames are stored as zero-padded JPEG images in individual subdirectories under `frames/`, one folder per student.

The script `photo2faces.py` then processes each frame using an OpenCV Haar cascade classifier loaded with

`cv2.CascadeClassifier(cv2.data.harcascades + "haarcascade_frontalface_default.xml")`. Faces are detected via `detectMultiScale()` with `scaleFactor=1.1` and `minNeighbors=5`. Each detected bounding box is expanded by 50 percent using

`new_x = max(0, x - (new_w - w)//2)` and related boundary checks so that the crop remains within the image size. The resulting cropped facial images are saved to a folder named `faces_cropped_50%/`. The dataset has not been explicitly mentioned in this report, as it is not publicly available and has been constructed using a private dataset.

5.2 Preprocessing

The script `p2_224x224.py` standardizes all cropped faces to a resolution of 224×224 pixels using `cv2.resize()`. This size is compatible with many convolutional neural networks, including VGG16, ResNet, and the dlib face recognition model utilized in this project. Images that fail to load are ignored after a `None` check.

Next, `convert2rgb.py` inspects the channel count of each image via `img.shape[2]` and converts any 4-channel BGRA images to 3-channel BGR using `cv2.cvtColor(img, cv2.COLOR_BGRA2BGR)`. Such BGRA images typically arise from PNG screenshots with transparency, which can cause errors when dlib attempts to compute encodings. A validation script, `validate_dataset.py`, iterates through all images in `fixed_photos/`, runs `face_recognition.face_encodings()` on each, and reports the number of valid encodings per student, flagging students with fewer than ten encodings for re-recording.

5.3 Face Encoding

The script `recog_face.py` loads each preprocessed image through `face_recognition.load_image_file()`, which returns an RGB NumPy array. This conversion is required because dlib expects RGB format, while OpenCV's `imread()` yields BGR. The function `face_recognition.face_encodings(rgb)` passes the image through dlib's pre-trained CNN to generate a 128-dimensional vector that captures the distinctive geometric characteristics of the face. All embeddings and their associated student names are stored in two parallel lists and then serialized into `encodings1.pkl` using Python's `pickle.dump()`. This file acts as the reference database for both entry and exit recognition modules.

5.4 Entry Camera: Part 1

The script `F_Recog.py` connects to the entry camera using `cv2.VideoCapture(1)` and automatically falls back to index 0 if index 1 is unavailable. For every captured frame, a BGR-to-RGB conversion is done with `cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)` before passing the frame to `face_recognition.face_locations()` and `face_recognition.face_encodings()`. For each detected face, `face_recognition.face_distance()` computes Euclidean distances between the live embedding and all stored embeddings. The index of the best candidate is obtained with `np.argmin(face_distances)`. A match is accepted only if `matches[best_match_index]` is `True` and the corresponding distance is less than 0.5.

Recognised students are stored in an attendance dictionary along with their entry timestamps, and duplicate entries are avoided by checking whether the name is already present in the dictionary. When the session is closed using the `q` key, the dictionary is exported as `attendance_YYYYMMDD_HHMMSS.csv` using the pandas library.

5.5 Exit Camera and Duration Verification: Part 2

The script `F_Recog_exit.py` opens the exit camera through `cv2.VideoCapture(0)`. It locates the latest entry CSV file with `glob.glob("attendance_*.csv")` combined with `max(csv_files, key=os.path.getctime)`. The same pipeline of face detection, RGB conversion, encoding, and distance-based matching with a 0.5 threshold is applied at the exit. Exit times are stored in an `exit_log` dictionary while ensuring that each student's exit is recorded only once.

After the exit session concludes, the script iterates over all names present in the entry CSV and calculates the stay duration.

$(\text{exit_time} - \text{entry_time}) \cdot \text{total_seconds()} / 60$. Students whose duration is at least 40 minutes are labeled as present, whereas those with shorter durations are marked absent. Three main cases are explicitly handled: students recognized at both cameras with a duration of at least 40 minutes (Present), students recognized at both cameras but with a duration below 40 minutes (Absent), and students recognized only at entry but not at exit (Absent, with exit time recorded as "Not recorded"). The final consolidated report is saved as `final_attendance_YYYYMMDD_HHMMSS.csv`.

6. System Flow Diagram

The end-to-end workflow of the system can be summarised as the following sequence.

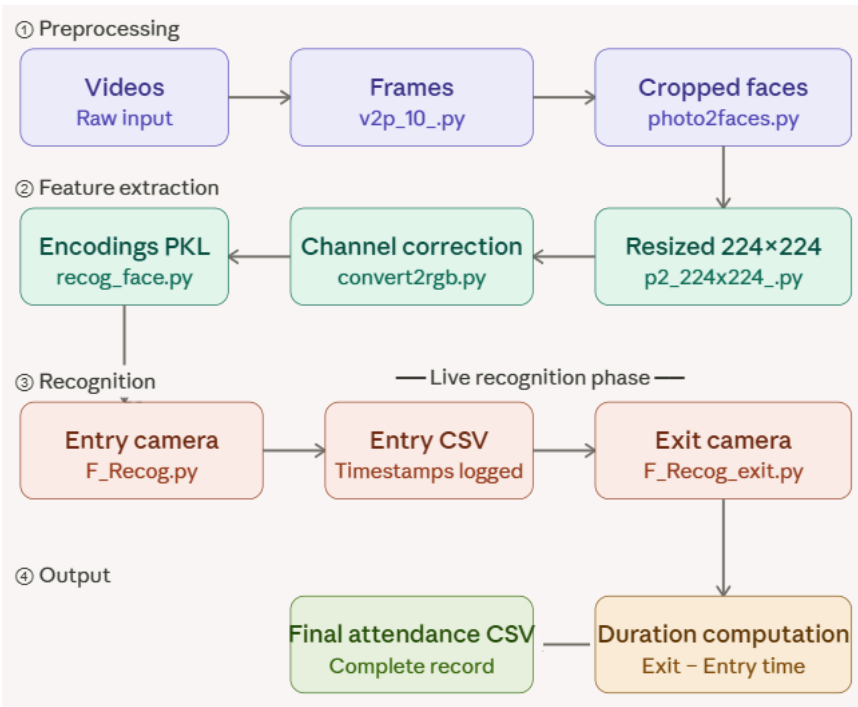


Fig 1. A single diagram illustrating this full pipeline should be included in the paper.

7. Result Analysis

7.1 Result Discussion

The system was evaluated with a group of N enrolled students over a simulated classroom session. The entry camera consistently recognized students as they walked into the room and stored their corresponding timestamps. The exit camera later identified the same students and recorded their departure times, enabling accurate duration calculations.

A sample of the final attendance CSV is shown below, where each row includes the student's name, date, entry time, exit time, total time in minutes, and status.

Name	Date	Entry Time	Exit Time	Duration (min)	Status
palash	25-03-2026	20:39:25	20:51:39	12.2	Absent
mangal	25-03-2026	20:39:36	20:52:02	12.4	Absent
ishita	25-03-2026	20:39:39	20:52:01	12.4	Absent

During testing, the 0.5 confidence threshold successfully filtered out a number of unknown faces that appeared at the entry camera, preventing them from being wrongly included in the attendance list.

7.5 Comparison With Existing Systems

The proposed system was qualitatively compared with manual attendance, a single-camera face recognition system, and an RFID-based system. Key differences are summarized below.

- Manual methods do not prevent proxy attendance, cannot verify duration, lack a confidence threshold, and do not support automatic CSV export, though they have no hardware cost.
- A single-camera face recognition system provides partial proxy prevention and low hardware cost, but still cannot validate how long students stay, lacks explicit distance-based thresholds, and handles false positives poorly.
- RFID-based systems prevent proxy and can automate CSV export, but require dedicated hardware tags, resulting in higher cost, and still do not capture the duration of presence.
- The proposed dual-camera approach delivers biometric proxy prevention, duration verification, confidence-threshold-based matching, automatic CSV output, low hardware cost, and better handling of false positives.

Feature	Manual Attendance	Single Camera FR System	RFID Based System	Your System
Proxy prevention	No	Partial	Yes	Yes
Duration verification	No	No	No	Yes
Confidence threshold	No	No	N/A	Yes
Automatic CSV export	No	No	Yes	Yes
Hardware cost	None	Low	High	Low
False positive handling	N/A	Poor	N/A	Good

Among the compared options, this system is the only one that simultaneously enforces biometric identity, ensures a minimum stay time, and filters uncertain matches through a distance threshold, making it well-suited for real classroom deployments.

8. Conclusion

This paper presented a face recognition–based attendance system that overcomes three major limitations of existing approaches: manual effort, proxy attendance, and the inability to verify the duration of presence. The design implements a complete pipeline, beginning with video capture and face detection, followed by preprocessing, CNN-based embedding generation, dual-camera recognition, and automated CSV export of attendance records. By applying a Euclidean distance threshold of 0.5, the system records only high-confidence matches. The requirement that students be recognized at both entry and exit with at least 40 minutes between events ensures that brief visits do not count as full attendance. The setup needs only two conventional cameras and no additional specialized hardware, making it practical and cost-effective. A current constraint is that performance can degrade under poor lighting, which can negatively impact detection and recognition accuracy.

9. Future Scope

Several enhancements can extend the functionality of the proposed system. A web-based dashboard could allow instructors to view, filter, and download attendance data by date, subject, or section. Integration with institutional ERP platforms would enable automatic synchronization of attendance with official academic records. Adding liveness detection at the entry camera

would help defend against photo or video spoofing attempts. The architecture can be scaled to multiple classrooms by using a central server that aggregates CSV files from different rooms. Finally, incorporating infrared cameras would allow reliable operation in low-light or nighttime conditions.

References

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